Fairness in AI-Assisted Emergency Response Systems: A Conditional Disparity Analysis of NYC EMS Dispatch

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Abstract

This study evaluates fairness in AI-assisted emergency dispatch by examining whether response outcomes differ across demographic groups for comparable emergency scenarios. Using 2024 NYC EMS incident data and 2022 ACS demographic proxies, we conduct a severity-stratified audit of response times. Our analysis employs Welch's t-tests and multivariate OLS regression to control for medical urgency. We find that response times significantly increase in correlation with the percentage of Black residents in a ZCTA, even when holding severity constant. This suggests that "fairness-blind" optimization in emergency systems may perpetuate systemic geographic disparities.

Keywords

fairness in AI, machine learning, algorithmic bias, emergency response, NYC OpenData, MAUP

1 Introduction

Emergency response systems operate in high-stakes, time-critical environments where the margin between life and death is often measured in seconds. In major metropolitan areas like New York City, the management of emergency response systems has shifted toward data-driven paradigms, with algorithmic decision-support tools increasingly used for dispatch prioritization, ambulance allocation, and route optimization. While these technologies aim to maximize system-wide efficiency, they raise profound questions regarding *algorithmic fairness*: do these "neutral" optimizations produce unequal outcomes for marginalized demographic groups?

This study investigates whether the emergency response process—and the opaque algorithmic systems that support it—produces disparate outcomes across demographic groups when medical scenarios are comparable in severity. We move beyond simple aggregate parity, which often masks underlying operational complexities, to focus on **conditional fairness**. This framework asks whether "like-for-like" emergencies receive equitable treatment regardless of the neighborhood's racial or socioeconomic composition.

Evaluating fairness in public-sector AI presents a methodological challenge because dispatch and triage algorithms are typically proprietary and inaccessible for direct inspection. Consequently, this project adopts a *proxy modeling* approach. By utilizing the NYC EMS Incident Dispatch dataset, we reconstruct the decision-making landscape to examine how system outputs—such as response times and transport decisions—correlate with neighborhood-level protected attributes. Following prior fairness auditing research, this approach evaluates disparities in system outcomes even when the underlying algorithm cannot be directly observed [Barocas et al. 2023].

In practice, emergency dispatch in New York City is supported by Computer-Aided Dispatch (CAD) systems, which assist with call intake, prioritization, and unit allocation. These systems are complemented by computerized triage tools used by the FDNY to support call classification and prioritization [Fire Department of the City of New York (FDNY) nd]. Together, these technologies reflect how rule-based and data-driven decision-support mechanisms are embedded within real-world emergency response workflows.

Our analysis centers on a severity-stratified audit. By grouping incidents by clinical priority and dispatch characteristics, we isolate the "service gap" from legitimate medical prioritization. We link city-level administrative data with American Community Survey (ACS) demographic proxies at the Zip Code Tabulation Area (ZCTA) level. This spatial lens allows us to identify potential "geographic penalties" and "time taxes" imposed on specific communities.

The contribution of this work is twofold. First, it provides an empirical, evidence-based assessment of real-world service disparities in a major urban center. Second, it demonstrates a robust framework for auditing opaque public-sector AI using open-source data. By distinguishing between operational necessities (such as traffic and call volume) and systemic demographic biases, we aim to provide a roadmap for more equitable, transparent, and accountable emergency dispatch systems.

2 Related Work

This research integrates three primary scholarly domains: formal definitions of algorithmic fairness, the auditing of opaque public-sector AI, and the empirical study of healthcare delivery disparities.

2.1 Formal Frameworks for Algorithmic Fairness

Foundational literature in machine learning fairness establishes that "fairness through blindness"—the mere removal of protected attributes—is insufficient to prevent bias [Barocas et al. 2023]. Hardt et al. [2016] introduced the concept of *equalized odds*, emphasizing that a model's performance must be balanced across groups when conditioned on the ground truth (in our case, clinical severity). In settings where the underlying model is proprietary or "black-box," Mehrabi et al. [2021] suggests that "outcome-based auditing" provides a robust mechanism to detect *disparate impact* without requiring direct access to model parameters.

2.2 Proxy Bias and Spatial Inequity

A critical risk in public-sector AI is *proxy bias*, where seemingly neutral variables correlate with protected characteristics. Obermeyer et al. [2019] famously demonstrated this in healthcare, where using

"healthcare spending" as a proxy for "medical need" led to systemic under-prediction of risk for Black patients.

In the context of emergency services, geographic variables often act as proxies for race and class. Research into "automated inequality" [Eubanks 2018] suggests that when systems optimize for efficiency using historical data, they risk codifying "spatial neglect"—the legacy of under-investment in specific neighborhoods. While Adam et al. [2022] and Visave and Doe [2024] have called for transparency in disaster-response AI, there remains a significant gap in the literature regarding how these theoretical biases manifest in real-time EMS dispatch metrics.

2.3 Empirical Disparities in Emergency Medical Services

Sociological and medical literature has long documented that emergency wait times are not uniform. Wang et al. [2024] identified significant racial disparities in emergency department throughput, attributing these to systemic hospital resource constraints. Within the specific field of EMS, studies have shown that neighborhood-level socioeconomic status can predict response latency [Sun and Chen 2022].

However, existing empirical studies often treat these as isolated sociological phenomena rather than *algorithmic failures*. Our work bridges this gap by applying the *conditional fairness* framework to administrative EMS data. By explicitly controlling for clinical severity, we transition the conversation from "general neighborhood differences" to a formal audit of whether the system maintains *predictive parity* across diverse urban geographies.

3 Data

This study uses two primary datasets. The first dataset is the NYC EMS Incident Dispatch dataset, which contains detailed operational records of emergency calls, including call type, dispatch priority, incident location, and multiple response time components. For this analysis, we use incident data from 2024 Q1. The second dataset is the American Community Survey (ACS) 2022 five-year estimates, which provide demographic and socioeconomic information at the Zip Code Tabulation Area (ZCTA) level.

Because the EMS dataset does not contain individual-level demographic attributes, we link EMS incidents to neighborhood-level demographic information using ZIP codes and construct area-level demographic proxies.

Two key variables are derived:

- **High Black ZIP:** A binary indicator identifying ZCTAs where the Black population percentage exceeds the citywide median.
- **Income Quartiles (*income_q*):** A categorical variable segmenting ZCTAs into four tiers based on Median Household Income.

After filtering for valid response times and removing invalid observations, the dataset contains several hundred thousand EMS incidents across all boroughs of New York City.

Because demographic variables are measured at the neighborhood level rather than the individual level, all findings should be interpreted as area-level associations rather than individual-level effects.

4 Methodology

The objective of our methodology is to isolate potential demographic disparities from operational factors that legitimately affect emergency response times. Our workflow follows a structured sequence: data integration, clinical stratification, and a multi-tiered statistical evaluation designed to handle the unique distributions of urban emergency data.

4.1 Data Integration and Preprocessing

To construct an analysis-ready dataset, we integrated NYC EMS dispatch records with neighborhood-level demographic data from the American Community Survey (ACS). EMS ZIP codes were standardized and mapped to Census Bureau Zip Code Tabulation Areas (ZCTAs), allowing each incident to be linked with ACS demographic indicators.

The EMS and ACS datasets were then merged using the standardized ZIP/ZCTA identifiers, providing socioeconomic context for each emergency incident while preserving operational variables such as call type and response time.

Following the merge, we applied basic data quality filters. Incidents with invalid or missing response time values were removed, including cases where the response-time validity indicator was not marked as valid or where response times were negative.

Finally, demographic proxy variables derived from the ACS data—including the High Black ZIP indicator and the income quartile variable (*income_q*)—were retained for use in the fairness analysis.

The resulting dataset combines incident-level EMS operational data with neighborhood-level demographic context, enabling conditional comparisons of emergency response outcomes across communities.

4.2 Severity Tiering and Clinical Controls

To evaluate *conditional fairness*—ensuring that "like-for-like" emergencies are compared—we stratified incidents into three clinically ordered tiers. This prevents the analysis from being confounded by different medical priorities across neighborhoods:

- **Tier 1 (High):** Life-threatening incidents (Severity Codes 1–2).
- **Tier 2 (Medium):** Urgent but non-imminent threats (Codes 3–4).
- **Tier 3 (Low):** Non-urgent or stable assignments (Codes 5–8).

4.3 Inferential Statistical Framework

To assess whether response outcomes differ significantly across demographic groups, we employed a dual-test strategy to address the non-Gaussian nature of EMS dispatch timing.

4.3.1 Welch's T-Test for Heteroskedastic Data. Preliminary Levene's tests indicated that the variance in response times differed significantly across geographic regions (violating the assumption of homoscedasticity). Consequently, we utilized Welch's t-test, which is robust to unequal variances and unequal sample sizes. The test

statistic t is defined as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where the degrees of freedom are calculated using the Welch–Satterthwaite equation.

4.3.2 Mann-Whitney U Test for Distributional Robustness. Emergency response data is characteristically right-skewed, frequently containing extreme "long-tail" outliers due to severe traffic or infrastructure delays. To ensure our findings were not driven solely by these outliers, we utilized the non-parametric Mann-Whitney U test. This test evaluates the null hypothesis that it is equally likely that a randomly selected observation from one group will be less than or greater than an observation from a second group, focusing on rank-order rather than raw means.

4.4 Multivariate Regression and Control Logic

To quantify the specific impact of protected attributes while controlling for environmental and operational factors, we estimated a series of Ordinary Least Squares (OLS) regression models. The primary model specification is formulated as:

$$Y_{rt} = \beta_0 + \beta_1 D_{race} + \beta_2 D_{income} + \gamma C_{ops} + \epsilon$$

Where:

- Y_{rt} represents the response time in seconds.
- D_{race} and D_{income} are the protected demographic proxies.
- C_{ops} is a vector of operational controls, including `Borough` (proxy for infrastructure), `Time_of_Day` (proxy for traffic), and `Severity_Tier`.

In addition to binary indicators such as High Black ZIP, the regression model also includes continuous demographic variables (e.g., Black Percent, Hispanic Percent, White Percent), which represent the proportion of each demographic group within a ZCTA, expressed on a 0–1 scale. As a result, regression coefficients for these variables should be interpreted as the effect of a full-unit increase (i.e., a change from 0 to 1 in population share), rather than a one-percentage-point increase.

4.5 Robustness and Multiple Testing Corrections

To mitigate the risk of Type I errors (false positives) inherent in testing multiple severity tiers and demographic indicators, we applied the Benjamini–Hochberg False Discovery Rate (FDR) correction to all p-values. Furthermore, we addressed the Modifiable Areal Unit Problem (MAUP) by conducting sensitivity checks at both the ZIP and Borough levels, ensuring that the direction and significance of our coefficients were not artifacts of spatial aggregation.

5 Results

5.1 Conditional Outcome Analysis

The primary objective of our initial analysis was to evaluate whether disparities in EMS response times persist when conditioning on the clinical severity of the incident. Table 1 presents summary statistics for incident response times (in seconds), stratified by the neighborhood racial proxy (High Black ZIP).

Table 1: Conditional Summary of Response Times (Seconds) by Severity and Racial Proxy

Severity Tier	High Black ZIP	n	Mean	Median	Std. Dev.
High (Tier 1)	0	42,358	444.34	384.00	313.16
	1	44,212	479.64	416.00	344.62
Medium (Tier 2)	0	67,175	587.96	459.00	495.46
	1	57,474	649.05	502.00	554.34
Low (Tier 3)	0	74,571	827.40	587.00	796.93
	1	77,693	959.82	668.00	914.62

A consistent trend emerges across all clinical categories: neighborhoods identified as High-Black ZCTAs experience systematically longer response times. For High-severity incidents involving life-threatening emergencies, the mean response time is 35.3 seconds longer in High-Black ZIP codes. This "service gap" widens significantly as clinical urgency decreases, with mean differences reaching 132.4 seconds for Low-severity calls. The parallel shift in both mean and median values suggests a structural delay in the distribution rather than the influence of isolated outliers.

5.2 Quantifying the Disparity Gap and Statistical Significance

To determine if these observed differences are statistically robust, we performed Welch's t-tests (to account for heteroskedasticity) and Mann-Whitney U tests (to account for right-skewed distributions). To mitigate Type I errors from multiple comparisons, we applied the Benjamini-Hochberg False Discovery Rate (FDR) correction.

Table 2: Significance Tests and Magnitude of Disparity Gap

Severity Tier	Mean Gap (s)	Median Gap (s)	Welch's p (adj)	MW-U p (adj)
High (Tier 1)	+35.30	+32.00	$< 10^{-50}$	$< 10^{-50}$
Medium (Tier 2)	+61.09	+43.00	$< 10^{-100}$	$< 10^{-100}$
Low (Tier 3)	+132.42	+81.00	$< 10^{-199}$	≈ 0.0

Note: All results are significant at the $p < 0.001$ level after FDR correction.

The results provide strong evidence of systematic disparities in response times across demographic groups. Even for the most critical calls, the probability of the disparity being due to chance is statistically negligible. Notably, the "time tax" imposed on marginalized neighborhoods is inversely related to medical urgency; as the city-wide system becomes less "strained" by emergency prioritization, the underlying geographic and demographic inequalities become more visible.

5.3 Distributional Variance and Interaction Trends

To visualize the consistency of these delays, Figure 1 displays the response distributions across severity tiers. To highlight the central tendencies and interquartile ranges (IQR), extreme outliers were omitted from the visualization.

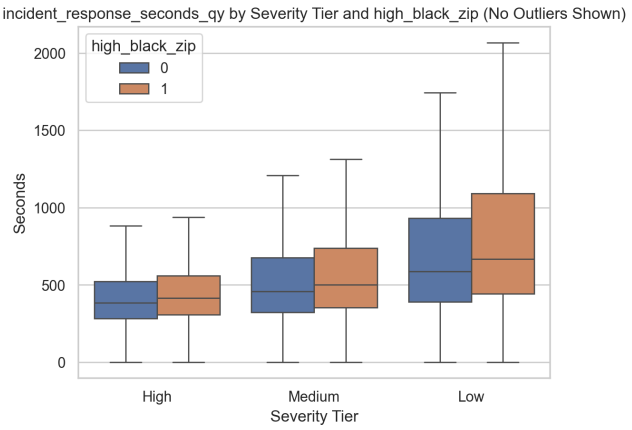

Figure 1: Response Time Distributions by Severity and Racial Proxy (Outliers Omitted)

The boxplot confirms that the entire distribution—not just the mean—is shifted upward in High-Black ZIP codes. This is further supported by the interaction plot in Figure 2, which illustrates the diverging slopes of response times. The non-overlapping 95% confidence intervals demonstrate that the disparity is not only significant but scales upward as a function of lower clinical priority.

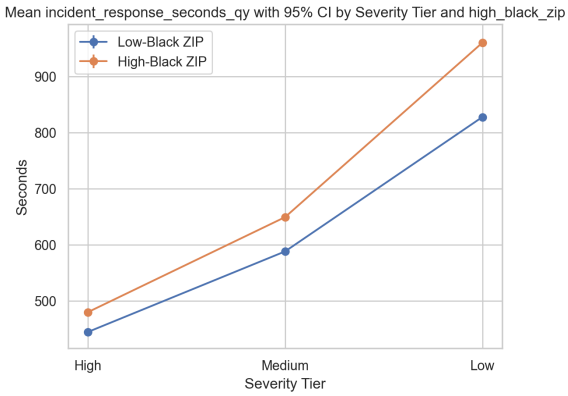

Figure 2: Divergence of Mean Response Times with 95% Confidence Intervals

5.4 Socioeconomic Intersections

We further investigated the coupling of economic status and service outcomes by stratifying response times by Median Household Income quartiles (Q1–Q4).

The ANOVA results ($p < 0.001$) confirm that neighborhood wealth is a significant predictor of dispatch efficiency. Residents in the lowest-income areas (Q1) wait, on average, 140 seconds longer for lower-priority calls than those in the highest-income areas (Q4). This fanning-out effect, visualized in Figure 3, reinforces the conclusion that operational friction—potentially due to lower ambulance density or higher relative call volume—is most acutely felt in socioeconomically disadvantaged ZCTAs.

Table 3: Socioeconomic Significance: Income Quartiles and Response Gaps

Severity	ANOVA (p)	Welch's p (Q1 vs Q4)	Gap (Q1-Q4)
High	3.97×10^{-20}	2.54×10^{-19}	+29.80s
Medium	8.31×10^{-72}	3.91×10^{-40}	+59.19s
Low	2.79×10^{-218}	2.73×10^{-102}	+140.15s

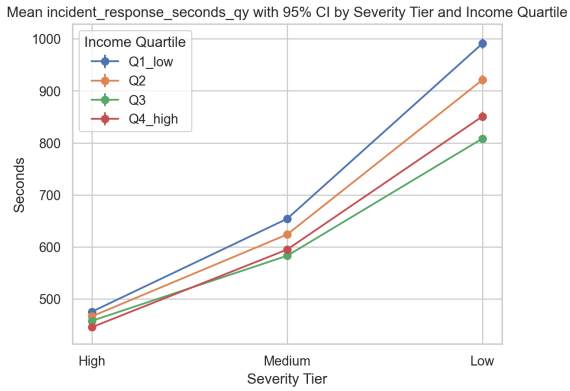

Figure 3: Mean Response Times by Severity Tier and Income Quartile

5.5 Multivariate Fairness Audit

To isolate the independent impact of race and income, we utilized a tiered OLS regression strategy. First, we established a **Baseline Control Model** (Table 4) to capture non-protected operational variance, such as traffic (Time of Day), infrastructure (Borough), and priority (Severity).

Table 4: Baseline OLS Results: Operational and Temporal Controls

Variable Group	Coefficient Range (β)	Avg. Std. Error	P-value
Intercept (Base)	399.20	6.72	< 0.001
Peak Traffic (4PM-6PM)	+356.00 to +411.55	7.30	< 0.001
Staten Island	-220.12	5.63	< 0.001
Severity: Low	+435.06	2.72	< 0.001

We then introduced demographic predictors to the model. The **Full Fairness Model** (Table 5) demonstrates that even when holding all operational factors (traffic, borough, severity) constant, the racial and economic composition of a neighborhood remains a statistically significant predictor of delay.

As shown in Table 5, the regression identifies a "demographic penalty": a full-unit increase in the proportion of Hispanic or Black residents (i.e., a change from 0 to 100% of the population) corresponds to an average delay increase of 193.92 and 143.34 seconds, respectively. In practical terms, a 1 percentage-point increase in these proportions corresponds to approximately 1.94 and 1.43 seconds of additional delay. Conversely, an increase in the White population percentage is associated with a decrease in response time ($\beta = -32.90$). These marginal effects are visualized in Figure 4,

Table 5: Full OLS Model: Isolated Impact of Protected Attributes

Predictor Variable	Coefficient (β)	Std. Error	P-value
High Black ZIP (Indicator)	+24.19	4.45	< 0.001
Median Income	+0.0001	4.87e-05	0.0002
Black Percent	+143.34	12.03	< 0.001
Hispanic Percent	+193.92	10.07	< 0.001
White Percent	-32.90	11.85	0.0055

Note: Percent variables are scaled 0-1. Coefficients represent change from 0% to 100%.

where the separation from the null-effect line (red dashed line) provides clear visual evidence of systemic disparity.

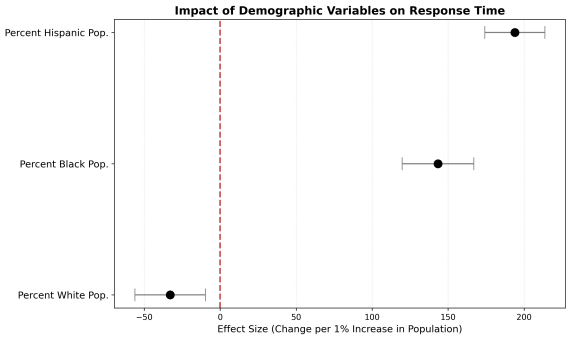

Figure 4: Marginal Effects of Demographic Variables on Response Time

5.6 Robustness Check: Scale Invariance (MAUP)

To ensure these findings were not artifacts of ZCTA-level spatial aggregation, we re-estimated the model at the **Borough level** (Table 6). This sensitivity check confirms that the relationship between racial composition and response delay is scale-invariant. While the coefficient magnitudes shifted due to the higher level of aggregation, the direction and significance ($p < 0.001$) remained stable. This cross-scale consistency (visualized in Figure 5) suggests that the disparities are not localized ZIP-code anomalies but are embedded in the city's macro-level emergency infrastructure.

Table 6: MAUP Sensitivity Check: Full Borough-Level OLS Model

Predictor Variable	Coefficient (β)	Std. Error	P-value
High Black Borough (Indicator)	+140.79	2.15	< 0.001
Median Income (Borough)	+0.0021	5.80e-05	< 0.001
Black Percent (Borough)	+40.94	0.68	< 0.001
Hispanic Percent (Borough)	+104.94	1.85	< 0.001
White Percent (Borough)	-28.76	0.82	< 0.001

Note: Results confirm that demographic predictors remain significant across scales.

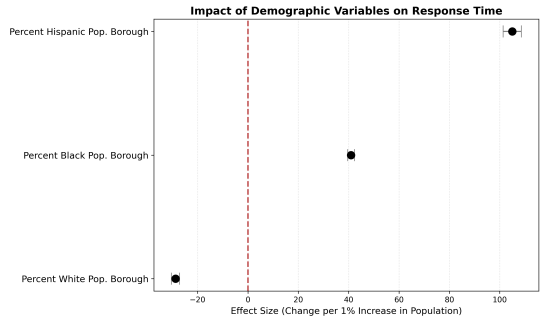

Figure 5: MAUP Sensitivity Check: Borough-Level Coefficient Plot

6 Discussion

The results of this audit provide empirical evidence of a "service gap" in New York City's emergency response system that aligns with existing racial and socioeconomic fault lines. While the EMS system is designed to be a universal safety net, our multivariate analysis reveals that geography and demographics act as significant modifiers of dispatch efficiency. These results suggest that efficiency-focused optimization in emergency response systems may unintentionally reproduce existing geographic inequalities.

6.1 The "Priority Paradox" and Structural Friction

Our analysis identified an inverse relationship between clinical urgency and the magnitude of disparity. While Tier 1 (High-severity) calls showed the narrowest gap (+35.3s), Tier 3 (Low-severity) calls exhibited a delay of over two minutes (+132.4s) in High-Black ZCTAs. This suggests a "Priority Paradox": while the system successfully "triage-adjusts" to minimize inequity during life-threatening events, the underlying structural friction—such as lower ambulance density, hospital deserts, or reduced transit infrastructure in marginalized neighborhoods—becomes the dominant factor in all other scenarios.

6.2 Proxy Bias and the Vicious Cycle of Efficiency

The persistence of these coefficients in our OLS model, even after controlling for borough and traffic, suggests the presence of *proxy bias*. Dispatch algorithms or resource allocation strategies optimized solely for "average efficiency" may inadvertently penalize neighborhoods with historically slower transit times. If historical data reflects a legacy of under-investment in certain ZCTAs, an AI-driven system that seeks to minimize travel time may continue to station fewer units in those areas, thereby risking the reproduction of historical service inequalities in future operational decisions.

6.3 Scale Invariance and the MAUP Check

The robustness of our findings across both ZCTA and Borough scales (addressing the Modifiable Areal Unit Problem) is particularly significant. Because the "demographic penalty" remains statistically significant at the macro-level, we can conclude that these delays are

not localized anomalies caused by specific street-level obstructions, but are rather systemic features of the city's macro-operational landscape.

7 Limitations

Several limitations should be acknowledged. Demographic attributes are measured at the ZIP-code level rather than at the individual level, which introduces potential ecological inference issues. In addition, the dataset does not include all operational variables that may influence response times, such as real-time ambulance availability or hospital capacity. Finally, the analysis is observational and therefore cannot establish causal relationships between demographic characteristics and response delays.

Future research could incorporate additional operational variables or evaluate policy interventions designed to reduce geographic disparities in emergency response.

8 Conclusion and Policy Recommendations

This study demonstrates that emergency response times in New York City vary systematically across neighborhoods with different demographic and socioeconomic characteristics. Even when accounting for the logistical challenges of urban density and peak-hour traffic, residents in High-Black and low-income neighborhoods face a statistically significant "time tax" for emergency services.

8.1 Algorithmic Equity Constraints

To move toward a more equitable system, municipal dispatch models must move beyond "efficiency-only" optimization. We recommend the integration of **Equity-Aware Constraints** that prioritize resource distribution based on neighborhood-level vulnerability indices rather than historical transit averages alone.

8.2 Infrastructure and Hospital Proximity

The widening gap in lower-severity tiers suggests that "turnaround time" at hospitals may be a significant latent variable. Policymakers should investigate the correlation between hospital closures in marginalized boroughs and the resulting increase in "out-of-service" time for local ambulances.

8.3 Final Summary

By quantifying the independent impact of race and income on life-saving services, this audit provides a roadmap for more transparent and equitable public safety operations. Achieving true service parity requires acknowledging that "neutral" operational data often carries the weight of historical disparity; only through intentional, equity-focused intervention can the city ensure that a resident's ZIP code does not determine their chance of survival in a crisis.

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